MBZUAI · Innovation Program 2026

# KDRIFT

*Automated Linux kernel driver patching — from 30-day CVE exposure to same-day protection*


PRINCIPAL INVESTIGATOR

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THEME

Cybersecurity & Digital Infrastructure

PITCH DATE

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# The problem worth solving

*What real-world pain exists today — and why current solutions fall short*

## THE PROBLEM

1. *The Linux kernel ships every 9–10 weeks, and each release breaks the internal APIs out-of-tree drivers depend on: 3,529 kernel CVEs in 2024 (10× the prior year), 8–9 new ones daily in 2025.*
2. *For UAE infrastructure: e&'s private 5G, ADNOC, the national cloud, every transition means manual work: reading changelogs, reverse-engineering API diffs, hand-writing patches.*
3. *Each transition takes 1–4 weeks, leaving a 30–60 day window where critical systems stay unpatched.*

***And Linux underpins nearly all of it: telecom, cloud, energy, government: every operator inherits this exposure.***

## WHY EXISTING APPROACHES FALL SHORT

1. *Manual patching does not scale: 8–9 CVEs/day across dozens of drivers is impossible for any engineering team.*
2. *Generic LLM tools (Copilot, Cursor) operate at file level with no kernel subsystem semantics, no API dependency graphs, and no compile-loop validation.*
3. *Kernel LTS backports only cover in-tree code; out-of-tree drivers (the real attack surface for UAE infrastructure) are entirely ignored.*
4. *The cost of failure: 1–4 weeks of senior kernel engineer time per transition, repeated every 9–10 weeks, with active exploitation of the resulting vulnerability window by nation-state actors.*

# Our solution

*What you're building — in a way a non-specialist can understand*

## WHAT YOU ARE BUILDING

*KDRIFT is an AI-automated pipeline for Linux kernel driver co-evolution.*
*We are building KDRIFT for telecom OEMs and infrastructure operators so they can receive validated kernel driver patches within hours of a CVE disclosure, not weeks.*

*The system works in three stages: (1) DETECT: a two-layer API drift detector builds an Interface Dependency Graph and identifies every breaking kernel API change; (2) SYNTHESIZE: a multi-agent LLM pipeline generates candidate patches, validates compilation, and runs smoke tests automatically; (3) DEPLOY: the patch set is delivered as a CI/CD step on every new kernel tag push.*

### WHO IT'S FOR

*Telecom OEMs (Nokia, Ericsson), semiconductor vendors (NVIDIA, Qualcomm, MediaTek),* ***UAE sovereign infrastructure operators*** *(e&, du, ADNOC), and national CERTs / regulatory bodies*

### WHAT THEY DO TODAY

*Manual kernel migration: 1–4 weeks of senior engineering time per kernel LTS transition, per driver subsystem, repeated every 9–10 weeks*

### WHAT CHANGES WITH US

*Push a new kernel version tag → KDRIFT detects all breaking API changes, synthesizes validated patches, and delivers a reviewed patch set in hours. Patch latency drops from 30–60 days to same-day.*

# What's truly novel

*The innovation — what you can do that nobody else can, and why*

**Assessed: INNOVATION · 25%**

## THE INNOVATION — IN 2-3 BULLETS

*1. Interface Dependency Graph (IDG): First formalization of API drift detection as a two-layer decidable problem: structural AST diff + LLM semantic classifier. Constructs a symbol-level dependency map between driver code and kernel internals. No prior work does this.*
*2. GRPO-Trained Patch Synthesis (AUTODRIVER): First RL-trained system specifically for kernel-level code evolution. Multi-agent architecture with reward modeling on 15+ years of kernel commit history. Generic LLMs fail to compile on kernel patching tasks at this level of specificity.*
*3. DRIFTBENCH: First structured benchmark for automated driver migration. Six complexity levels, 500+ real kernel version transitions. Creates a scientific evaluation standard where none previously existed.*
*4. Verified Patch Acceptance: We integrate an agentic formal-verification stage built on AProver (github.com/agentic-prover/aprover).*

## CLOSEST EXISTING APPROACHES — AND HOW YOU

*Baseline 1: GitHub Copilot / Cursor*

*Baseline 2: KernelGPT (arXiv 2024)*
*Our edge: Targets kernel bug finding, not driver co-evolution. No patch synthesis pipeline.*

*Baseline 3: Manual BSP engineering (current industry practice)*
*Our edge: Does not scale to 8–9 CVEs/ day across multiple driver stacks.*

# Commercial opportunity

*Market, customers, and how this becomes a self-sustaining asset*

**Assessed: COMMERCIAL VIABILITY 30%**

## TARGET CUSTOMER

*Telecom OEMs (Nokia, Ericsson), RAN BSP maintenance teams who must re-validate out-of-tree drivers on every kernel LTS upgrade. Semiconductor vendors (NVIDIA, Qualcomm, MediaTek), driver release engineering teams. UAE sovereign infrastructure operators (e&, du, ADNOC), security and compliance teams under TDRA mandate to patch CVEs within 30 days. National CERTs and regulatory bodies including the UAE Cyber Security Council (verbal interest confirmed).*

***++ send to UAE Cyber Security Council Youcheng's new paper***

## MARKET SIZE

*TAM: $4.2B, global embedded Linux BSP services market (Grand View Research, 2025). SAM: $680M, enterprise kernel maintenance tooling for telecom and cloud OEMs. SOM: $42M, Year-3 target covering UAE + GCC region + top-5 global RAN vendors. Driver: every 9–10 week kernel release creates a mandatory re-engagement with every customer.*

## BUSINESS MODEL & PRICING

*Year 1 — Paid pilot contract: AED 200–500K/year per anchor customer. Includes integration, SLA, and quarterly reports. Year 2+: SaaS subscription: $2,500–8,000/month per active driver repository monitored (self-serve CI/CD). Enterprise license: $150–400K/year for vendors with 50+ driver stacks (NVIDIA, Qualcomm), including KERN model fine-tuning on proprietary codebase.*

## PATH TO REVENUE

*First paying customer: UAE Cyber Security Council or MediaTek (both at verbal interest stage). Use case: automated driver patch delivery for their Linux infrastructure stack. To close: deliver a working demo on their own driver codebase during Month 5 validation phase. Target: pilot contract signed by Month 6, first SaaS subscription revenue by Month 12.*

# Competitive advantage & IP

*Why a follower cannot trivially copy what you build*

**Assessed: IP & DEFENSIBILITY · 10%**

## WHAT MAKES IT HARD TO COPY

*Data moat: DRIFTBENCH is the only structured dataset for kernel driver migration, built from 15+ years of kernel commit history — replicating it takes months. Model performance gap: KERN-7B/32B fine-tuned on kernel-specific code outperforms generic LLMs (GPT-4o, Gemini) on compile-valid patch generation. Switching cost: once a vendor integrates KDRIFT into their CI/CD and we have ingested their driver codebase, migration to a competitor requires rebuilding the integration from scratch. Domain know-how: 18 months of accumulated kernel subsystem expertise cannot be hired quickly.*

## IP STRATEGY

*Patent filings planned: (1) Interface Dependency Graph construction method for kernel API drift detection; (2) GRPO-based reward shaping for kernel-level code synthesis. KERN model checkpoints: MBZUAI IP — open-weight release for academic use, commercial license for enterprise customers. DRIFTBENCH: open dataset with MBZUAI attribution, creating community lock-in around our evaluation standard. Trade secret: proprietary training data curation pipeline and GRPO reward function design.* ***(3) ++ prof Youcheng prover work be a patent***

## OWNERSHIP & COMMERCIAL ROUTE

*IP owned by MBZUAI. Commercialization route: spinout or exclusive license to a UAE-incorporated entity. Industry partners currently at verbal interest stage (UAE Cyber Security Council, MediaTek). Path from prototype to deployment: PoC → paid pilot with anchor customer → SaaS platform → spinout with MBZUAI equity participation.*

# Market & impact

*The reach of this work — strategically, economically, and societally*

**Assessed: MARKET & IMPACT · 15%**

### UAE STRATEGIC FIT

*UAE National AI Strategy 2031 and UAE Cybersecurity Strategy both prioritize sovereign digital infrastructure resilience. KDRIFT directly addresses TDRA's mandate for operators to patch critical CVEs within 30 days, automating compliance. The e& × ADNOC private 5G network (largest single-site globally) runs on Linux with out-of-tree drivers. UAE Cyber Security Council has confirmed verbal interest in the project as a sovereign patch velocity tool.*

### ECONOMIC IMPACT

*Telecom OEM savings: estimated $2–4M/year per vendor in BSP engineering costs eliminated. Patch latency reduction: 30–60 days to hours directly closing the active exploitation window. Potential AED 500M+ in avoided UAE infrastructure disruption costs over 5 years. New industry category: automated kernel maintenance as a service, with recurring SaaS revenue scalable across every kernel release cycle globally.*

### END-USER IMPACT

*Infrastructure operators: no more emergency patching sprints after CVE disclosure. UAE citizens: more resilient 5G, cloud, and government digital services. Security teams: automated patch delivery frees engineers to focus on policy rather than plumbing. Research community: DRIFTBENCH and KERN model released as open scientific assets from MBZUAI, establishing the university as a global reference point for AI-for-systems research.*

# Team & track record

*Why this team will execute — credentials, experience, partners*

**Assessed: TEAM & EXECUTION · 15%**

## CORE TEAM

| Name | Role | Why they're the right person |
|---|---|---|
| *Chun Jason Xue* | *PI / Professor* | *Systems & security expert; SUF grant holder* |
| *Youcheng Sun* | *Co-PI / Asst. Prof.* | *Program analysis & formal verification; co-supervisor on KDRIFT* |
| *Arina Kharlamova* | *MSc, incoming PhD* | *Lead engineer* |
| *Jiawen Liu / Humaid Alqasimi / Tianyi Zhang / Xinrui Yang ++Ivaylo Ivanov* | *PhD & MSc Researchers* | *Pipeline dev (Jiawen); DRIFTBENCH & data curation (Humaid, Tianyi, Xinrui)* |

## INDUSTRY / CLINICAL PARTNERS

*UAE Cyber Security Council, verbal interest confirmed; pilot discussion initiated.*
*MediaTek, verbal interest; driver dataset access under discussion.*
*Both to be converted to formal MoU during Month 5 of the project.*

## PRIOR EXECUTION OF SIMILAR WORK

*KDRIFT: 18 months continuous development; arXiv:2511.18924*
*SUF grant (Prof. Xue) already funding the team, this program accelerates commercialization.*

# project plan & milestones

*Concrete deliverables, the path to TRL 5–7, and what success looks like*

| Phase 1 · Months 1–2 | Phase 2 · Months 3–4 | Phase 3 · Month 5 | Phase 4 · Month 6 |
|---|---|---|---|
| KEY DELIVERABLES<br>DRIFTBENCH v1.0 dataset (500+ transitions)<br>Interface Dependency Graph for top-20 drivers<br>QEMU CI/CD validation environment live | KEY DELIVERABLES<br>AUTODRIVER multi-agent pipeline deployed<br>GRPO training complete; >70% patch success L1-3<br>~~NVIDIA + Qualcomm integration test~~<br>**(++Uae companies)** | KEY DELIVERABLES<br>*DRIFTBENCH + KERN model public release*<br>*SOSP 2026 paper submitted*<br>*2x patent filings*<br>*PoC demo ready for investors*<br>***++ International companies*** | KEY DELIVERABLES<br>*1 pilot contract signed*<br>*First SaaS subscription revenue target*<br>*Spinout pitch materials prepared* |

**Starting TRL → Ending TRL: TRL 2–3 → TRL 6 · Decision-quality outputs by project end date: pilot-ready PoC, DRIFTBENCH dataset, 2x IP filings, 1 pilot contract signed**

# The ask

*Budget, support requested, and what success looks like in 18 months*

**BUDGET BREAKDOWN — TOTAL AED 1,000,000**

*Personnel AED 500K · Compute AED 300K · IP filings AED 80K · Partner/pilot AED 70K · Dissemination AED 50K*

*Personnel builds the core pipeline. Compute enables GRPO training, not feasible on CPU. IP budget protects IDG + GRPO synthesis before public release. Partner budget unlocks NDA dataset access and first pilot.*

**WHAT WE NEED FROM MBZUAI BEYOND FUNDING**

*Industry introductions to e&, Nokia UAE, and TDRA (highest-leverage ask). IP support and MBZUAI commercialization framework. HPC cluster priority access for H100 training. NVIDIA/ Qualcomm NDA driver dataset access.*

**MEASURABLE SUCCESS AT MONTH 18**

*① DRIFTBENCH released ② KERN model released ③ SOSP 2026 submitted ④ 2 patents filed ⑤ 1 pilot contract signed*

***++ add a bit more strong***

*KDRIFT turns kernel CVE exposure from a 30-day liability into a same-day solved problem — for UAE infrastructure, and every Linux operator beyond.*